F24-113-D-TeleAI: AI-Conversational Agent for Restaurant Order Automation

Project Team

Ahmad Raza Zahid	20I-0853
Haseeb Ahmed	21I-2712
M.Sohail	21I-2720

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Supervised by

Dr. Hasan Mujtaba

Co-Supervised by

Mr. Saad Salman

Department of Computer Science

National University of Computer and Emerging Sciences
Islamabad, Pakistan

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Chapter 1

Introduction

Managing a restaurant requires careful balancing. It's understandable how things may easily get out of control when clients are coming in, phones are ringing nonstop, and orders are coming in from all directions. Envision yourself in the busiest time of day in a restaurant. The employees are running around trying to assist customers as the phone keeps ringing. On the other side, while you're working to serve the customers in front of you, you're losing out on business because of backed-up phone lines or employees who can't process orders fast enough.. Not only does this affect your present sales, but clients who receive sluggish service will never forget it.

Imagine now a solution that takes this off of your team and smoothly processes orders over the phone. This is the reason behind the design of \textbf{Tele-AI}. It acts as your smart assistant, handling reservations, streamlining phone ordering, and even remembering customer preferences and dietry constraints so that staff members can focus on what really matters—delivering an amazing dining experience. This means that during peak hours, restaurant owners will have less tension, happier customers, and better productivity .

1.1 Problem Statement

Assume you run a well-known restaurant. On this Friday evening, all of the tables are occupied, and calls for orders and reservations are coming in continuously. Despite their best efforts, your team is limited in what they can accomplish at once. Orders are overlooked, reservations are handled carelessly, and the phone is frequently busy when more people try to call because of their juggling act between taking phone calls and serving diners in the restaurant. It's an irritating scenario for your customers as well as for your team,and each missed call equals to a lost sale.

In today's fast-paced world, efficiency is key to staying competitive. But many restaurants are stuck with outdated processes. Managing phone orders and reservations manually leads to slow service, errors, and dissatisfied customers. Hiring more staff or relying on expensive technology isn't always feasible, especially for smaller businesses and startups.

This is where **Tele-AI** steps in. By automating phone orders and reservations, It helps restaurant owners streamline their operations, reduce staff workload, and increase sales by ensuring that no customer is left waiting on hold or frustrated by a busy phone line. In a competitive industry where customer experience is everything, our system offers a smarter and more efficient way to manage the chaos of phone-based orders and reservations, allowing restaurant owners to focus on growing their business.

1.2 Scope

The scope of the our project is to fully automate the process of handling orders and reservations for restaurants through voice-based interactions. The system will process customer phone calls, transcribe orders and reservations, and interpret the requests with minimal human intervention. Additionally, it will remember customers' preferences and dietary constraints from their previous interactions, ensuring personalized experiences for returning customers.

The system will include several key features:

- A web terminal for displaying real-time orders and reservations.
- Sentiment analysis to evaluate customer satisfaction during calls.
- Upselling and cross-selling by suggesting related premium items.
- The ability to make outbound calls to promote marketing offers and collect feedback.
- Analytics dashboard to provide restaurant owners and staff with insights into customer experiences and sales performance.
- Automates the reservation process, enabling customers to book tables via phone calls and get updated by phone or message notification.

Overall, **Tele-AI** will create an efficient and intelligent order management solution for restaurants.

1.3 Modules

The proposed project includes the following core modules:

1.3.1 Module 1: Automatic Speech Recognition (ASR)

Responsible for converting customer speech into text in real-time, enabling accurate transcription of orders and reservations.

- Real-time transcription of customer orders.
- High accuracy in recognizing customer requests.

1.3.2 Module 2: Natural Language Understanding (NLU)

Interprets customer queries and intents, enabling the system to understand and act upon spoken commands.

- Understands customer requests such as placing orders or making reservations.
- Identifies customer preferences and dietary constraints.

1.3.3 Module 3: Business Logic

Manages the flow of operations based on customer inputs and interactions with other modules.

- Processes orders and reservations.
- Handles upselling suggestions.
- Manages dietary constraints.

1.3.4 Module 4: Dedicated Terminal

A web app terminal displaying orders, reservations, and system analytics for restaurant staff and managers.

- Real-time display of incoming orders.
- Insights into customer satisfaction and system performance.

1.3.5 Module 5: Sentiment Analysis

Evaluates customer satisfaction based on factors like voice tone and call duration.

- Analyzes customer satisfaction during the call.
- Provides sentiment scores displayed on the dashboard.

1.3.6 Module 6: Outbound Marketing Calls

Automated system for promoting marketing deals and collecting customer feedback through outbound phone calls.

- Automated calls informing customers about special offers.
- AI-generated promotional messages.

1.3.7 Module 7: Reservation Management

Automates the reservation process, enabling customers to book tables via phone calls or an app.

- Manages reservations in real-time.
- Provides customers with available table options.
- Sends reservation confirmation to the customer.

1.4 User Classes and Characteristics

Identify the various user classes expected to interact with the **Tele-AI** system and describe their roles and characteristics.

User Class	Description
Restaurant Owners	Restaurant owners who want insights into customer's satisfaction, sales performance, and order management. They will use the analytics dashboard to evaluate performance and metrics about sales and also calls data.
Restaurant Staff	The staff will interact with the system to manage customer orders, reservations, and access real-time information via the dedicated terminal. They require minimal training to use the system and adopt to it..
Restaurant Managers	Managers will oversee the restaurant operations and ensure smooth workflow. They will analyze performance metrics and customer sentiment through the analytics provided by the system.
Customers	Customers who will place orders or make reservations via phone calls. Ai assistant will assist them by providing personalized services based on previous interactions, offering upsell suggestions, and also promoting deals.
Technical Team	Developers and IT staff responsible for maintaining and enhancing the system. They will ensure the system runs smoothly, troubleshoot any issues, and improve system features.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the TeleAI project.

2.1 Event Response Table

The following are event-response tables for the project. These tables focus on the backend processes of an AI-driven restaurant order management system.

2.1.1 Event Response Table

Event	Trigger	Response
Customer initiates AI conversation	Customer speaks via drive-thru or phone call	AI bot responds and takes the order
Order confirmation	AI finishes taking customer order	AI sends confirmation to both kitchen and customer
Dietary restriction added	Customer mentions dietary restrictions	AI bot logs restriction and adjusts order according to that
Customer requests upselling recommendation	Customer asks for recommendations	AI Assistant suggests menu items based on past behavior or current order
Order update	Customer modifies order during the conversation	AI updates the order and informs both kitchen and customer
Third-party delivery request	Customer selects delivery	AI processes delivery request and forwards details to third-party services
Order canceled	Customer cancels order before confirmation	AI cancels the order and logs it as canceled
Feedback request after delivery	Order marked as complete by third-party delivery	AI requests customer feedback via voice or app
AI sentiment analysis on customer voice	AI detects emotion during interaction	AI adjusts its responses to create a positive customer experience
Table reservation request	Customer requests a reservation via voice	AI logs reservation, confirms availability, and sends details to customer
Safe menu browsing	Customer browses menu on his phone or call	AI ensures secure browsing and prevents potential phishing attempts

Table 2.1: Event Response Table

2.2 Functional Requirements

This section describes the functional requirements of the TeleAI system.

2.2.1 Module 1: ASR (Automatic Speech Recognition)

Purpose: Convert customer speech into text in real-time.
Features:

- Real-time speech transcription of customer's orders and reservations.
- High accuracy in speech recognition of customer requests and preferences.

Requirement ID	Description	Priority
FR1-001	The system shall provide real-time transcription of customer speech into text for order placement.	High
FR1-002	The system shall ensure high accuracy in recognizing customer speech for both orders and reservations.	High
FR1-003	The system shall ensure cleaning noise in both background and in network.	High

Table 2.2: Functional Requirements for ASR Module

2.2.2 Module 2: NLU (Natural Language Understanding)

Purpose: Interpret customer queries and intents.
Features:

- Understands customer requests, such as placing orders or making reservations.
- Identifies dietary constraints or special requests.

Requirement ID	Description	Priority
FR2-001	The system shall interpret customer queries to process orders and reservations.	High
FR2-002	The system shall identify special dietary constraints mentioned during the conversation.	High

Table 2.3: Functional Requirements for NLU Module

2.2.3 Module 3: Business Logic

Purpose: Manage the flow of operations based on customer input.
Features:

- Processes orders and reservations.
- Handles upselling suggestions based on the customer's order.
- Manages dietary constraints to ensure customer safety.

Requirement ID	Description	Priority
FR3-001	The system shall process customer orders and reservations.	High
FR3-002	The system shall provide upselling suggestions based on order content.	Medium
FR3-003	The system shall enforce dietary restrictions throughout the ordering process.	High

Table 2.4: Functional Requirements for Business Logic Module

2.2.4 Module 4: Dedicated Terminal

Purpose: Display orders, reservations, and insights for staff and managers.
Features:

- Real-time display of incoming orders and reservations.
- Provides access to insights and analytics, including customer satisfaction ratings.

Requirement ID	Description	Priority
FR4-001	The system shall display incoming orders and reservations in real-time for staff.	High
FR4-002	The system shall provide customer satisfaction analytics for managers.	Medium

Table 2.5: Functional Requirements for Dedicated Terminal Module

2.2.5 Module 5: Sentiment Analysis

Purpose: Rate customer satisfaction on a scale of 1 to 10.
Features:

- Analyzes calls based on factors like duration and tone.
- Provides sentiment scores for each call, visible on the analytics dashboard.
- Allows stakeholders to listen to call recordings for detailed reviews.

Requirement ID	**Description**	**Priority**
FR5-001	The system shall analyze customer sentiment based on voice tone and call duration.	Medium
FR5-002	The system shall provide a sentiment score for each customer interaction.	Medium
FR5-003	The system shall allow managers to review call recordings for performance monitoring.	Medium

Table 2.6: Functional Requirements for Sentiment Analysis Module

2.2.6 Module 6: Outbound Marketing Calls

Purpose: Inform customers about promotions and special offers.
Features:

- Automated outbound calls to inform customers about deals.
- Pre-recorded or AI-generated messages with promotional information.

Requirement ID	Description	Priority
FR6-001	The system shall perform outbound marketing calls to notify customers about promotions.	Medium
FR6-002	The system shall use pre-recorded or AI-generated messages for promotional campaigns.	Medium

Table 2.7: Functional Requirements for Outbound Marketing Calls Module

2.2.7 Module 7: Reservation System

Purpose: Automate the reservation process for customers.
Features:

- Allow customers to make reservations via voice command.
- Notify customers of confirmed reservations.
- Handle reservation cancellations or updates.

Requirement ID	**Description**	**Priority**
FR7-001	The system shall allow customers to make reservations via voice command.	High
FR7-002	The system shall notify customers of reservation confirmations.	High
FR7-003	The system shall allow customers to cancel or update reservations.	Medium

Table 2.8: Functional Requirements for Reservation System

2.3 Non-Functional Requirements

This section specifies the non-functional requirements for the TeleAI system, focusing on unique needs related to the problem and scope.

2.3.1 Reliability

- **NFR-01:** The system shall maintain a continuous and reliable connection with third-party delivery services, ensuring that no order data is lost during transmission.
- **NFR-02:** The system shall automatically switch to a backup server within 3-5 seconds in case of server failure, ensuring uninterrupted service.

2.3.2 Usability

- **USE-01:** The voice interface shall allow customers to modify or cancel an order seamlessly within 2 interactions.
- **USE-02:** The system shall enable restaurant staff to view, edit, or manually intervene in AI-driven orders from an intuitive dashboard in less than 2 clicks.
- **USE-03:** The AI system will provide clear, real-time feedback to users confirming the order status after each interaction.

2.3.3 Performance

- **PER-01:** The AI system shall process both inbound and outbound customer orders via phone in under 2 seconds.
- **PER-02:** The system shall analyze customer sentiment through voice tone recognition with 90% accuracy during interactions.

2.3.4 Security

- **SEC-01:** The system shall prevent data breaches through AI-based anomaly detection, identifying unauthorized access attempts in real-time.
- **SEC-02:** Access to customer order history and preferences shall be restricted based on user roles, with restaurant managers having more permissions than standard staff.

- **SEC-03:** The system shall log all interactions related to sensitive order information and retain logs for 30 days to assist in auditing.

2.4 Domain Model

The domain model provides a detailed overview of the Teleai project's structure and functionality. Here it illustrates the key components involved, their interactions, and the data flow between them.

Customer Data: The model represents customers with their unique identifiers, contact information, and order history. This data is used for various purposes, including order processing, reservation management, and targeted marketing.

Order Processing: The BusinessLogic component handles order processing, including item selection, pricing, and dietary constraint management. Orders are created, modified, and confirmed or canceled based on customer preferences and menu availability.

Reservation Management: This module manages reservations, including creating, modifying, and canceling reservations. Reservation details such as date, time, and number of people are captured and stored.

Customer Satisfaction: SentimentAnalysis component evaluates customer satisfaction based on call data. This information can be used to identify areas for improvement and enhance the overall customer experience.

Marketing Targeting: Marketing Calls part identifies suitable customers for promotions based on their data. This helps tailor marketing efforts to individual customer needs and preferences.

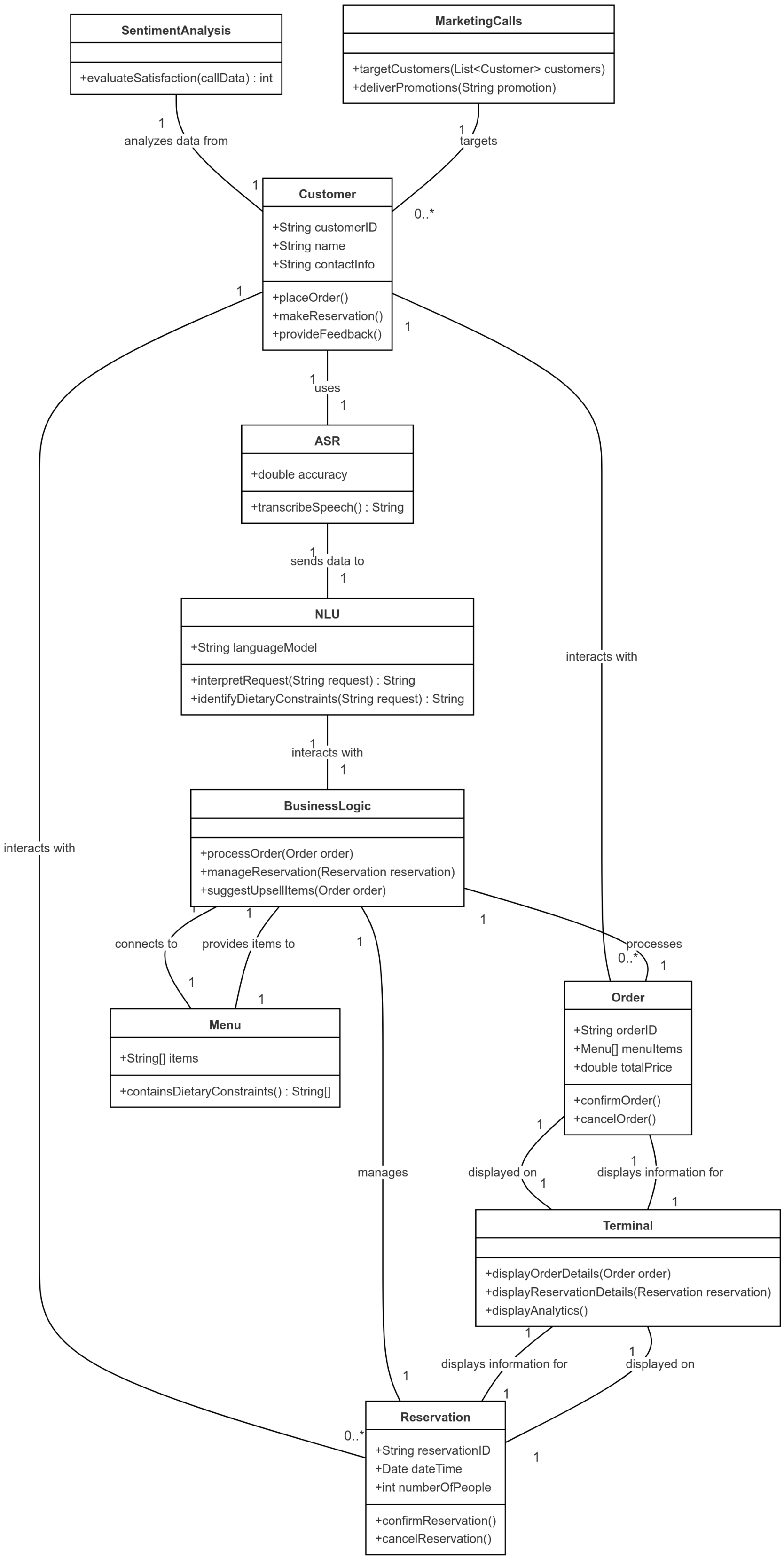


Figure 2.1: Domain Model for TeleAI System

Chapter 3

System Overview

The **Tele-AI** system is an AI-driven solution designed to automate the order management and reservation processes for restaurants. By integrating advanced technologies like Automatic Speech Recognition (ASR), Natural Language Understanding (NLU), and sentiment analysis, **Tele-AI** provides a seamless experience for customers and restaurant staff alike. The system acts as an intelligent virtual assistant capable of handling phone orders, managing reservations, and offering personalized recommendations, all while reducing the workload on restaurant staff.

The primary goal of the system is to streamline customer interactions, improve operational efficiency, and enhance customer satisfaction through automation. With minimal human intervention, **Tele-AI** manages orders, tracks customer preferences and dietary constraints, and provides managers with critical insights into the restaurant's performance via a dedicated analytics dashboard.

3.1 Context and Background

In many restaurants, managing phone-based orders alongside in-person customers can be a challenging task for staff. Often, this leads to longer wait times for customers, missed orders due to busy phone lines, and reduced efficiency in service. Our product addresses these challenges by automating the process of handling phone calls, allowing staff to focus on serving customers present in the restaurant.

Furthermore, the system's ability to recall customer preferences based on historical data allows for more personalized service, which is often difficult to achieve in busy environments. By also incorporating sentiment analysis,Our system provides restaurant managers with real-time feedback on customer satisfaction, enabling them to adjust services as needed.

3.2 General Functionality

The core functionality of our system includes:

- **Order Management:** Automatically transcribes and processes customer orders placed over phone calls. It captures the details of the orders accurately using ASR and passes them through NLU for understanding specific requests such as dietary constraints or modifications.

- **Reservation Management:** Automates the process of booking reservations for customers. It interprets requests, checks for available slots, and confirms reservations with minimal input from restaurant staff.

- **Sentiment Analysis:** During customer interactions, the system assesses the tone and mood of the customer, providing real-time sentiment scores. This data is crucial for gauging customer satisfaction and improving service quality.

- **Upselling and Cross-selling:** Based on the customer's order history and preferences, the system intelligently suggests additional items or premium products that complement their order.

- **Outbound Marketing Calls:** The system is capable of making automated marketing calls to inform customers of special promotions or collect feedback. It delivers AI-generated or pre-recorded messages, contributing to increased customer engagement.

- **Analytics Dashboard:** Provides restaurant managers with a detailed view of real-time orders, reservations, sentiment analysis, and other key metrics. This information helps them monitor operational performance and make data-driven decisions.

3.3 System Design

Tele-AI is designed with modular architecture, allowing each component to interact seamlessly while maintaining independent functionality. Key components of the system include:

- **ASR Module:** Converts spoken customer input into text for further processing.

- **NLU Module:** Interprets customer intent, ensuring accurate order processing and handling of complex requests.

- **Business Logic:** Handles all operational logic, including order processing, reservation management, and upselling suggestions.
- **Terminal:** Displays real-time orders, reservations, and analytics for restaurant staff, offering an easy-to-use interface.
- **Sentiment Analysis Module:** Assesses the tone and sentiment of customer interactions to measure satisfaction.
- **Outbound Marketing:** Generates calls to customers promoting special offers or seeking feedback.

These modules work together to create an efficient, AI-driven system that improves service quality while reducing the burden on restaurant staff. **Tele-AI** aims to offer a future-ready solution that not only enhances customer service but also provides restaurants with the tools they need to optimize operations and grow their business.

3.4 Architectural Design

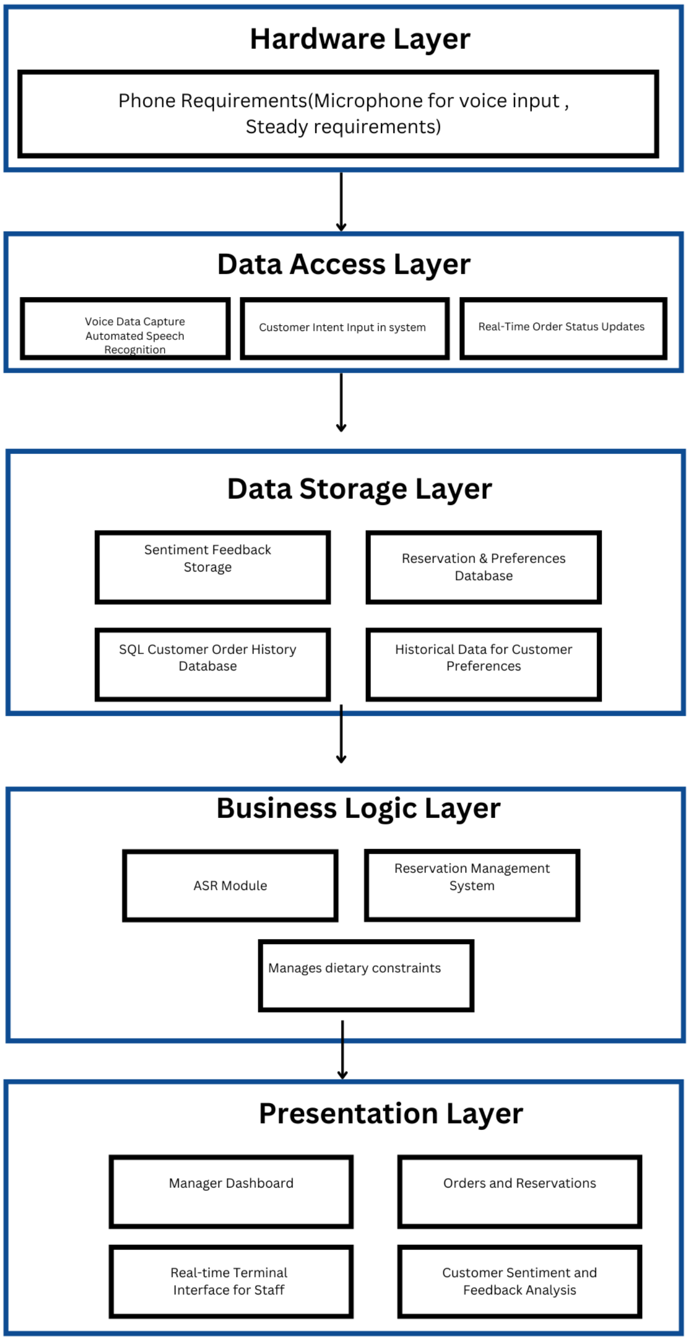
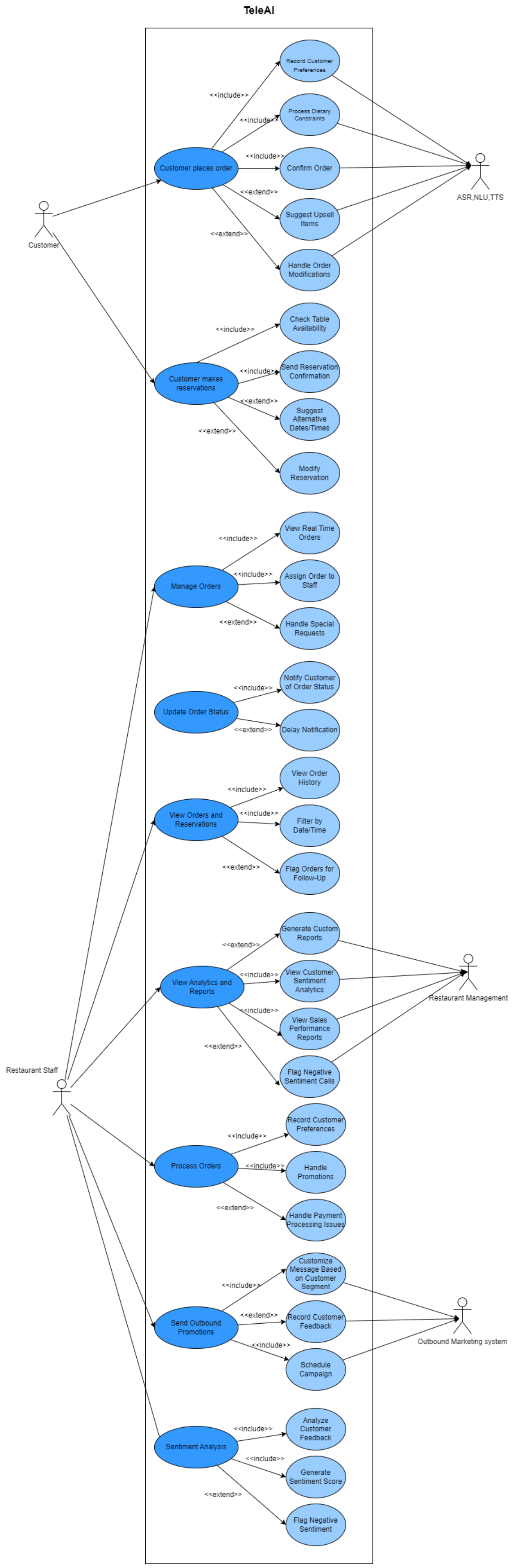


Figure 3.1: System Architecture Design

3.5 Design Models

Following are our Design Models which include the detailed Use-Case diagrams,Class Diagram, State Transition Diagram and Sequence Diagram.

Figure 3.2: Use Case Diagram



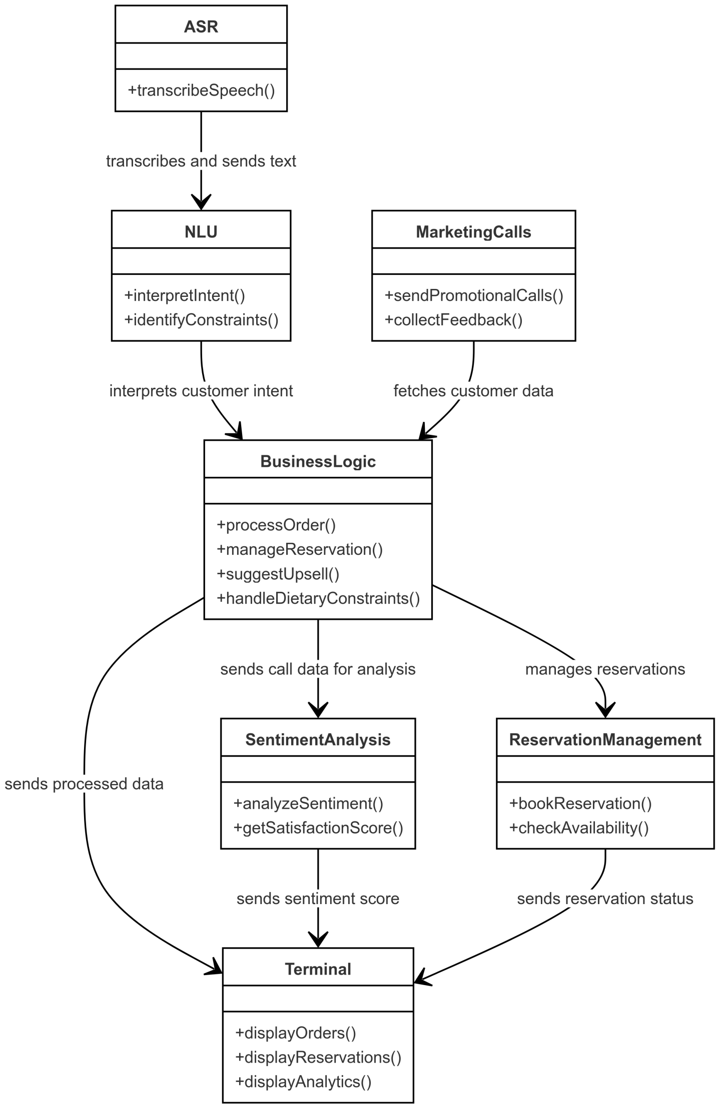


Figure 3.3: Class Diagram

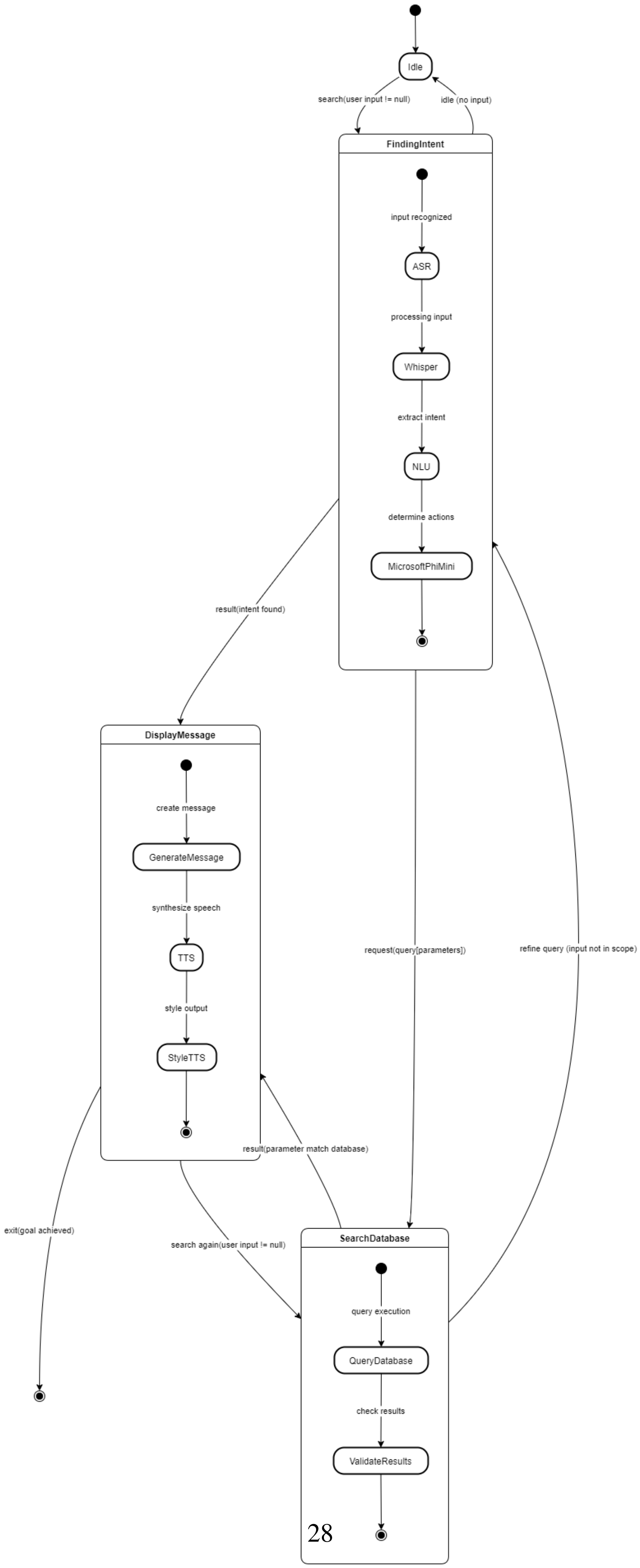


Figure 3.4: State Transition Diagram

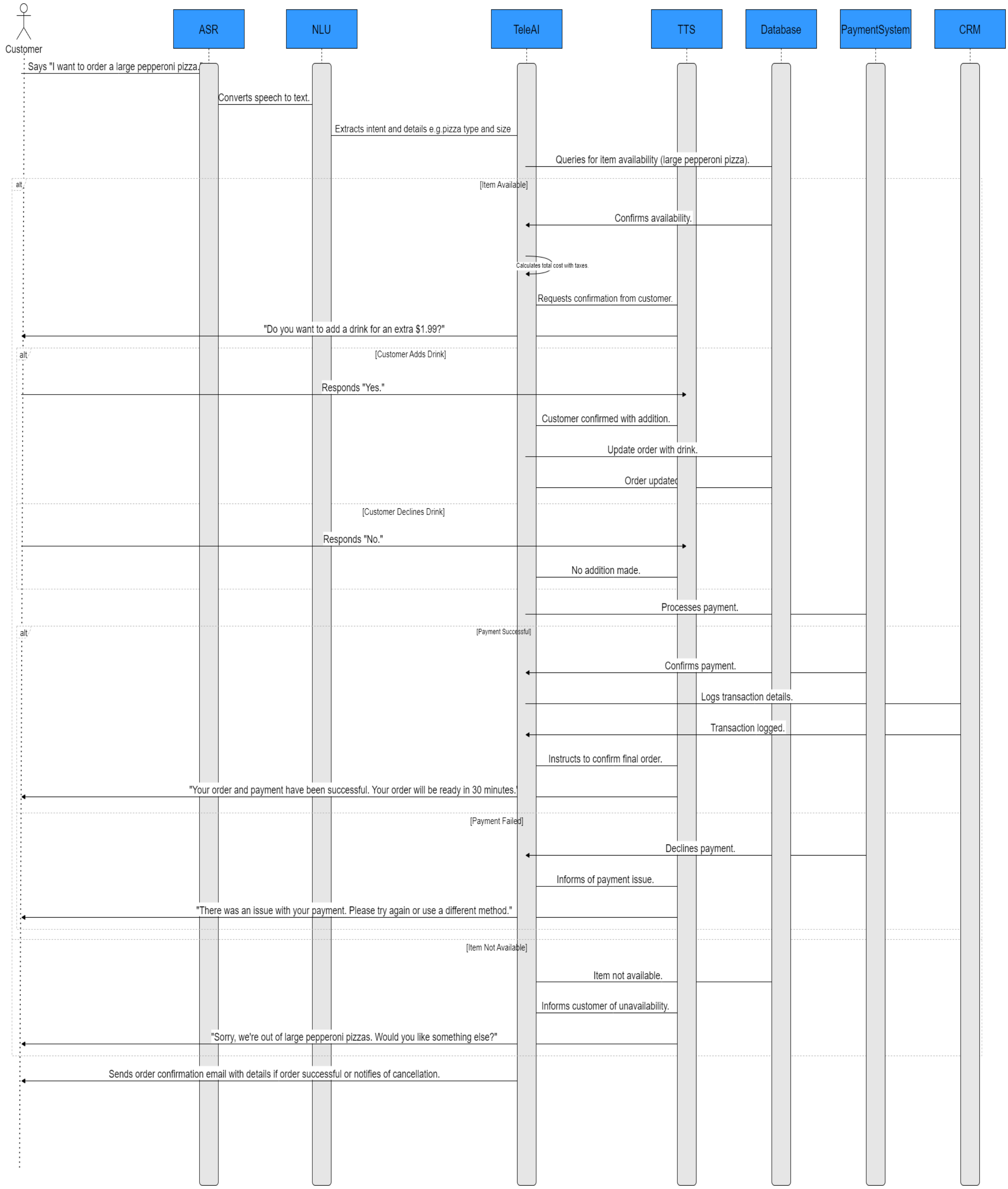


Figure 3.5: Sequence Diagram: Interaction during Order Processing

This Sequence Diagram provides a visual representation of the order processing workflow in the Tele-AI system. It outlines how different system components such as the ASR Module, NLU Module, and Business Logic interact to manage a typical order from initiation to confirmation. This diagram is crucial for understanding the real-time operations and data flow within the system.

3.5.1 Use Case Sequence Diagrams

3.5.2 UC-01: Place Order Sequence Diagram

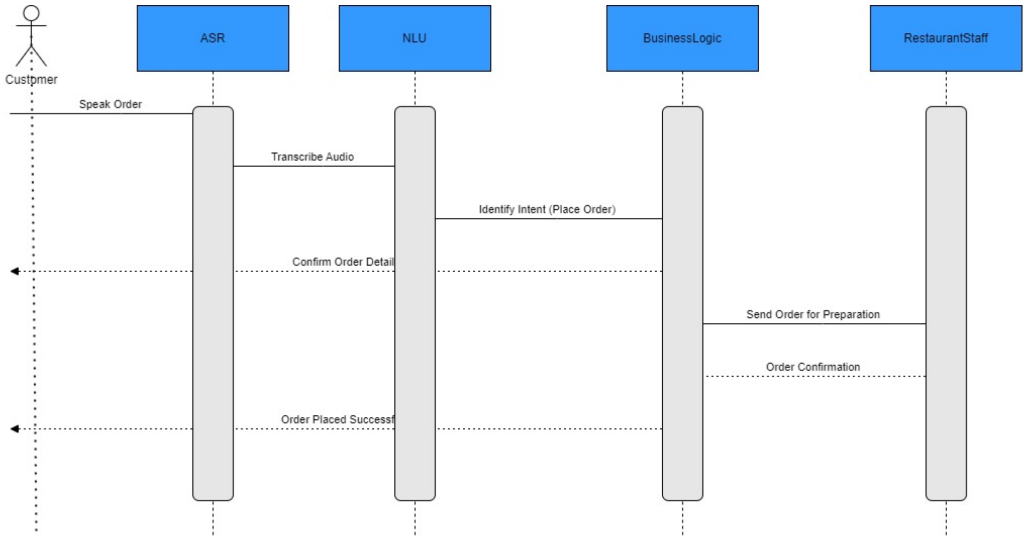


Figure 3.6: Place Order Sequence Diagram

3.5.3 UC-02:Manage Orders Sequence Diagram

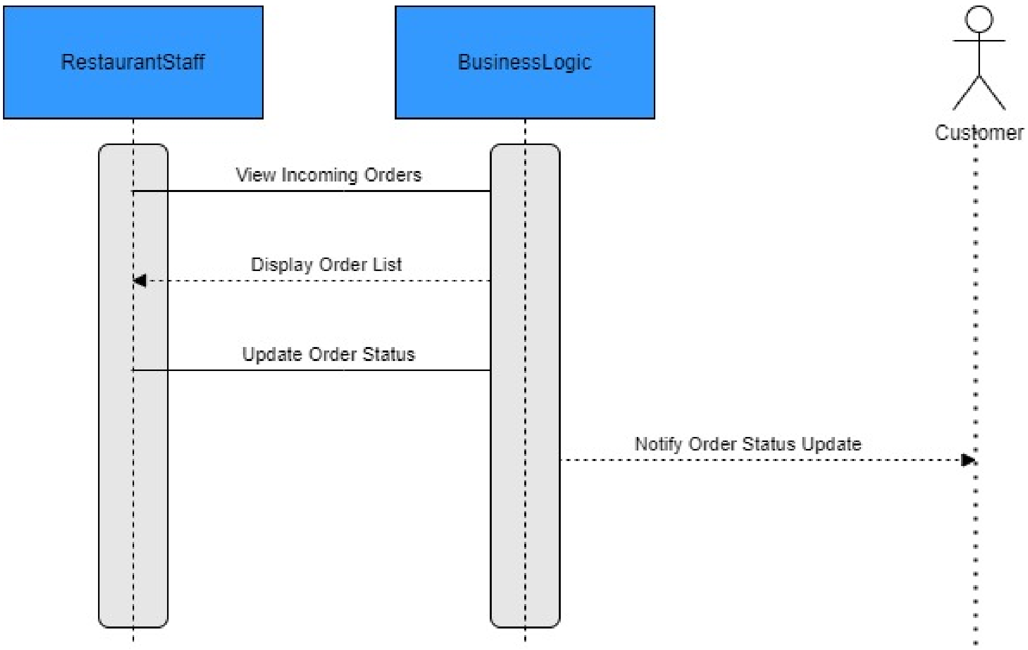


Figure 3.7: Manage Orders Sequence Diagram

3.5.4 UC-03:Make Reservation Sequence Diagram

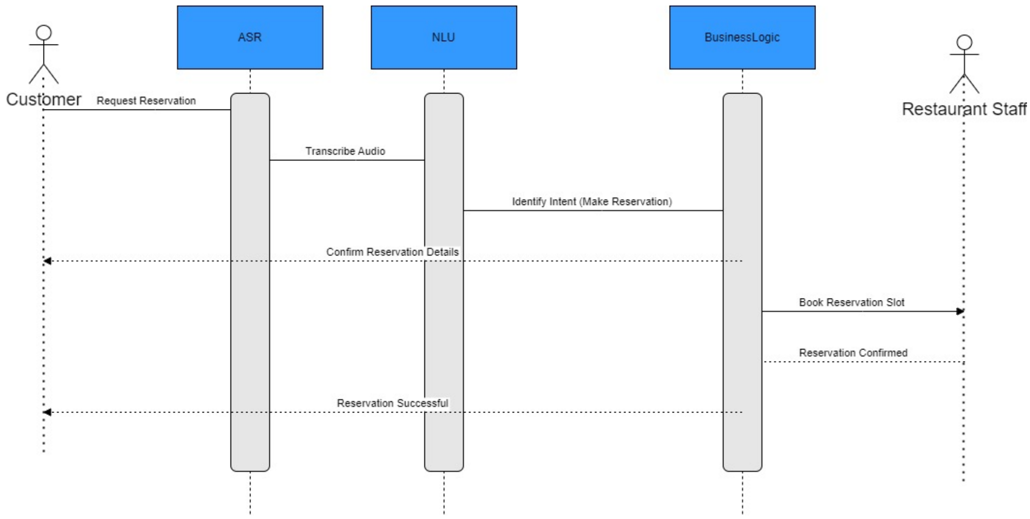


Figure 3.8: Make Reservation Sequence Diagram

3.5.5 UC-04:Sentiment Analysis Sequence Diagram

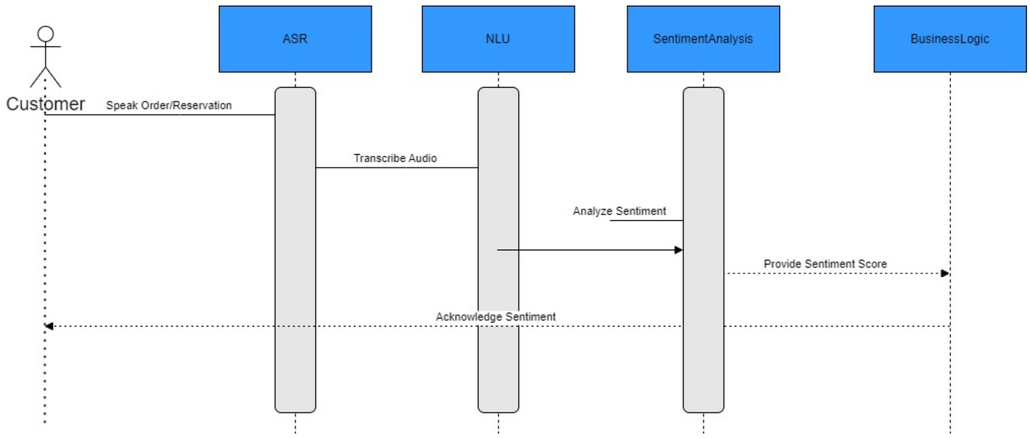


Figure 3.9: Sentiment Analysis Sequence Diagram

3.5.6 UC-05:Send Outbound Promotions Sequence Diagram

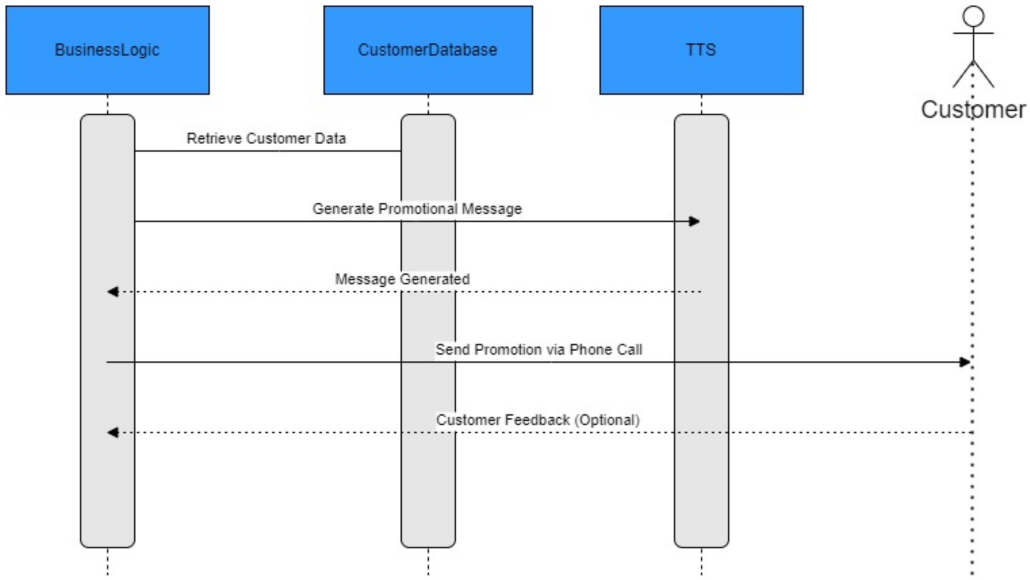


Figure 3.10: Send Outbound Promotions Sequence Diagram

3.6 Data Design

For the Tele-AI system, we've chosen MySQL as our primary data storage solution. MySQL is well-suited for managing structured data, which is essential for handling the various components of our restaurant order management system. The way we organize and store data reflects the core functionalities of our application, ensuring that everything is efficient and easy to access.

3.6.1 Major Data Entities

- **User Data:** We gather essential information from restaurant owners, including their name, email, and password. This data is crucial for creating accounts and managing user authentication. It's securely stored in a MySQL database, allowing for quick access and management.

- **Order Data:** Each restaurant can manage orders that come in via phone calls or through the app. This includes details like order items, total amounts, and customer preferences. Storing this information in MySQL helps in tracking and managing orders effectively.

- **Menu Data:** Restaurants need to maintain an updated menu, which includes item names, descriptions, prices, and availability. By storing menu data in MySQL, we ensure that updates are reflected in real-time, providing customers with accurate information.

- **Customer Feedback:** We collect customer feedback to enhance service quality. This data, including ratings and comments, is stored in a dedicated table within our MySQL database, allowing restaurant owners to analyze feedback and make improvements.

- **Order Tracking Data:** To provide real-time tracking of orders, we store data related to each order's status. This includes timestamps for order placement, preparation, and delivery. Utilizing MySQL for this allows us to manage and retrieve data efficiently for updating customers on their order status.

3.6.2 Data Flow and Processing

The data flow within our system is straightforward and logical. During user registration, we collect their information and store it securely in MySQL. This sets the stage for a seamless user experience, enabling restaurant owners to manage orders efficiently.

When an order is placed, relevant details are captured and stored in the order database. The system processes these orders in real-time, updating their status and notifying the relevant parties through the interface. Additionally, customer feedback is gathered and stored, helping restaurant owners to make informed decisions based on customer preferences.

The order tracking feature relies on efficient data processing. As orders move through various stages, timestamps are recorded, allowing us to keep customers updated on their order status. This real-time capability enhances the overall user experience.

3.6.3 Databases and Data Storage

The Tele-AI system utilizes a MySQL database to manage and store all essential data:

- **User Database:** This database stores user credentials and profile information, facilitating user management and secure authentication.
- **Order Database:** All order-related data is kept here, including order details, status updates, and tracking information.
- **Menu Database:** This database houses the menu items for each restaurant, ensuring that customers have access to the most current offerings.
- **Feedback Database:** Customer feedback is stored separately, allowing for easy analysis and tracking of service quality.

This structured approach to data design ensures that the Tele-AI application can effectively manage, process, and display information relevant to restaurant owners and their customers, ultimately providing a better ordering experience.

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